Clarinda Montilla

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I am a Masters Astronomy student at the University of Canterbury, interested in time-domain astronomy, fast transients, and sky surveys. My thesis focuses on how to improve detection methods in TESS as well as analysing transients for the TESSELLATE sky survey. I am eager to continue working with fast transients and hope to find a PhD beginning in 2026.

Education

March 2025 - April 2026	**M.Sc, University of Canterbury** Astronomy. Grade: Distinction Thesis title: *The Need for Speed: Modelling Fast Transients*. Supervisors: Dr. Ryan Ridden-Harper, Dr. Joscha Jahns-Schindler and Associate Professor Michael Albrow Key Skills: Machine learning, multiband photometry, open-souce collaborative pipeline development, programming, paper writing.
February 2024 - November 2024	**B.Sc Hons, University of Canterbury**, Astronomy. Grade: First Class Thesis title: *Finding the Fastest Explosions with TESS*. Supervisor: Dr. Ryan Ridden-Harper Key Skills: Data simulation, data recovery, photometry, transient science.
February 2021 – November 2023	**B.Sc, University of Canterbury**, Physics and Mathematics

Scholarships

2025	**Masters Scholarship funded by Royal Society of New Zealand, Te Apārangi Marsden Fund**, University of Canterbury
2024	**Summer Vacation Studentship**, Swinburne University of Technology
2023	**Summer Research Student**, University of Canterbury
2022	**Summer Observer at the Mount John Observatory**, University of Canterbury
2021	**Go Waitaha Canterbury Scholarship**, University of Canterbury **Hiranga Scholarship**, University of Canterbury

Publications

1. Z. G. Lane, R. Ridden-Harper, S. Rest, A. Rest, C. L. Ransome, Q. Wang, **C. Montilla**, *et al.*, "SN 2019vxm: A Shocking Coincidence between Fermi and TESS," *arXiv preprint arXiv:2511.15975*, 2026.
2. **C. Montilla**, J. Jahns-Schindler, *et al.*, "Beaming in fast radio bursts," In preparation, 2026.
3. H. Roxburgh, R. Ridden-Harper, A. Moore, **C. Montilla**, *et al.*, "TESSELLATE: Piecing Together the Variable Sky With TESS," *arXiv preprint arXiv:2502.16905*, 2025.

Contributed Talks & Posters

July 2026	**Contributed Talk** –Astronomical Society of Australia, Annual Scientific Meeting 2026; The Need for Speed - Modelling Fast Transients
February 2026	**Contributed Talk** – Transients in Middle Earth; The Need for Speed - Modelling Fast Transients

Contributed Talks & Posters (continued)

October 2025	**Contributed Talk** – FTSky2025; Beaming in Fast Radio Bursts
July 2025	**Contributed Talk** – FRB2025; Beaming in Fast Radio Bursts
	Contributed Talk – Astronomical Society of Australia, Annual Scientific Meeting 2025; Colouring High Cadence Light-curves of Optical Transients
	Poster – Astronomical Society of Australia, Annual Scientific Meeting 2025; Beaming in Fast Radio Bursts

Employment History

June 2023 – present

Townsend Teece Observer and Outreach Guide, University of Canterbury.

- Lead tour groups ranging from 10-20 people, through the Townsend Teece Observatory Tower and the University of Canterbury Ōtehīwai Mount John Observatory.
- Setup and pack down of the telescopes.
- Participate in outreach to primary schools through the Children's University program.

February 2024 – present

Astronomy Tutor, University of Canterbury.

- Tutor for ASTR109: The Cosmos: Birth and Evolution (Semester One) from 2024 to 2026.
- Organised and planned tutorial sessions for online and in-person classes (around 50 students).
- Graded and gave feedback for assignments and exams.

Skills

Coding	Python, R, LaTeX.
Data Analysis	Photometry, machine learning, multiband photometry, data simulation, pipelines, statistical analysis. Skilled in Word and Excel
Observing Experience	MOA 1.8m Telescope and B&C 0.6m telescope at University of Canterbury Mount John Observatory.

- Fully trained to use both MOA and B&C in January 2023.
- Service mode observing.
- Routine maintenance of telescopes.
- Gathered data for several different science projects (including LOOK, YSE, LSST Deep Drilling Fields, DEBASS).
- Filled out nightly logs to track observations, weather conditions and altitude of targets.
- Hosted several tour groups and visiting professors.
- Assisted in the re-aluminizing of the McLellan 1m mirror.

Outreach

As one of the key members of the University of Canterbury Astronomy Outreach group since 2023, I have participated in many outreach events, ranging from regular telescope tours, open days, to visiting primary schools. Below is a selection of some of the events I have run in the past year.

Matariki Events, University of Canterbury Outreach

- Setup, operation and pack down of C11 telescopes.
- Assisted in guided laser tours of the stars.

Children's University, University of Canterbury

- Setup, operation and pack down of the solar-scopes
- Organised educational games for children of ages 5–12, focused on developing an understanding of basic astronomical concepts.

Referees

Dr. Ryan Ridden-Harper, University of Canterbury
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Associate Professor Michael Albrow, University of Canterbury
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